Specialist Mathematics

Written Examination 1

2024 Insight Publications Trial Examination

- **Reading time:** 15 minutes
- **Writing time:** 1 hour

Approved materials

- Question and Answer book
- Students are permitted to bring into the examination room: pens, pencils, highlighters, erasers, sharpeners and rulers.
- Students are NOT permitted to bring into the examination room: any technology (calculators or software), notes of any kind, blank sheets of paper and /or correction fluid/tape.

Materials supplied

- Question and Answer book of 15 pages
- Formula sheet
- Working space is provided throughout the book.

Instructions

- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- All responses must be in English.

Students are **not** permitted to bring mobile phones and/or any unauthorised electronic devices into the examination room.

Contents

Questions (10 questions, 40 marks) _____ 2–12

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Instructions

- Answer **all** questions in the spaces provided.
- Unless otherwise specified, an **exact** answer is required to a question.
- In questions where more than one mark is available, appropriate working **must** be shown.
- Unless otherwise indicated, the diagrams in this book are **not** drawn to scale.
- Take the **acceleration due to gravity** to have magnitude g ms^{-2}, where g = 9.8.

Question 1 (3 marks)

Evaluate $\int_1^4 \sqrt{x}\log_e(2x)\,dx$.

Give your answer in the form $a\log_e(2)+b$, where $a,b \in R$.

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Question 2 (4 marks)

a. Evaluate $\left(1+\sqrt{3}i\right)^5+\left(1-\sqrt{3}i\right)^5$.

2 marks

b. Find the values of k for which $\left(1+\sqrt{3}i\right)^k+\left(1-\sqrt{3}i\right)^k=128$, where k is a positive integer.

2 marks

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Question 3 (3 marks)

Consider the curve defined parametrically by

$$x = 2\cos(t) + \cos(2t),\ y = 2\sin(t) - \sin(2t), \text{ where } 0 \le t \le 2\pi.$$

Find the arc length of the curve from $t = 0$ to $t = \frac{\pi}{3}$.

Question 4 (3 marks)

Consider the relation $x\cot(y) - 2y^2 = 8$.

Find the gradient of the curve at the point $\left(\frac{\pi^2}{8} + 8, \frac{\pi}{4}\right)$.

Give your answer in the form $\frac{a}{\pi^2 + a\pi + b}$, where $a, b \in R$.

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Question 5 (4 marks)

Use mathematical induction to prove that $\left(\cos\left(\theta\right)-\sin\left(\theta\right)i\right)^{n}=\cos\left(n\theta\right)-\sin\left(n\theta\right)i$ for all $n\in Z^{+}$.

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Question 6 (6 marks)

Consider the Cartesian equations of two planes:

$\prod_1 : x + 2y - z = 15$

$\prod_2 : x + y + pz = 5$

a. Given that $\prod_1$ and $\prod_2$ are perpendicular, show that $p = 3$.

1 mark

b. Find the Cartesian equation of the plane that is perpendicular to both $\prod_1$ and $\prod_2$, and which also passes through point $(1, 2, 1)$.

2 marks

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Line L, given by the equation $\underset{\sim}{r}(t) = \underset{\sim}{i} + \underset{\sim}{j} + \underset{\sim}{k} + t\left(\underset{\sim}{i} + 3\underset{\sim}{j} - 6\underset{\sim}{k}\right), t \in R$, intersects $\prod_1$ at point A and $\prod_2$ at point B.

c. Find the distance AB.

3 marks

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Question 7 (4 marks)

An ice cube tray with 6 cuboid compartments is filled with water and placed in a freezer to make 6 ice cubes. Each compartment has a 3 cm by 3 cm square base.

The height of each ice cube, H, is normally distributed with a mean of 2 cm and a standard deviation of $\frac{1}{6}\text{ cm}$. Assume that the height of each ice cube is independent.

a. Find the expected total volume of ice in the ice cube tray in cm^3.

1 mark

b. Find the standard deviation of the total volume of ice in the ice cube tray in cm^3.

Give your answer in the form $\frac{a\sqrt{b}}{c}$ where a, b and c are positive integers.

2 marks

c. Find the expected surface area of one ice cube in cm^2.

1 mark

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Question 8 (4 marks)

The position vector of a particle moving relative to the origin at time t seconds is given by $\underset{\sim}{r}(t) = 2\cos(t)\underset{\sim}{i} + 4\sin(t)\underset{\sim}{j} + 2t\underset{\sim}{k}$ for $t \geq 0$. All distances are measured in metres.

a. Find the maximum and minimum speeds of the particle.

2 marks

b. Find the number of times, between $t = 0$ and $t = 2\pi$, that the velocity is perpendicular to the acceleration.

2 marks

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Question 9 (5 marks)

A tank initially holds $200\ \mathrm{L}$ of water in which $16\ \mathrm{kg}$ of salt has been dissolved. Pure water then flows into the tank at a rate of $2\ \mathrm{L}$ per minute. The mixture is stirred continuously and flows out of the tank at a rate of $4\ \mathrm{L}$ per minute.

a. Show that the differential equation for x, the number of kilograms of salt in the tank after t minutes, is given by

$$\frac{dx}{dt} = -\frac{2x}{100-t}.$$

1 mark

b. Solve the differential equation given in **part a.** to find x as a function of t. Express your answer in the form $x = \left(\frac{a-t}{b}\right)^2$, where a and b are positive integers.

2 marks

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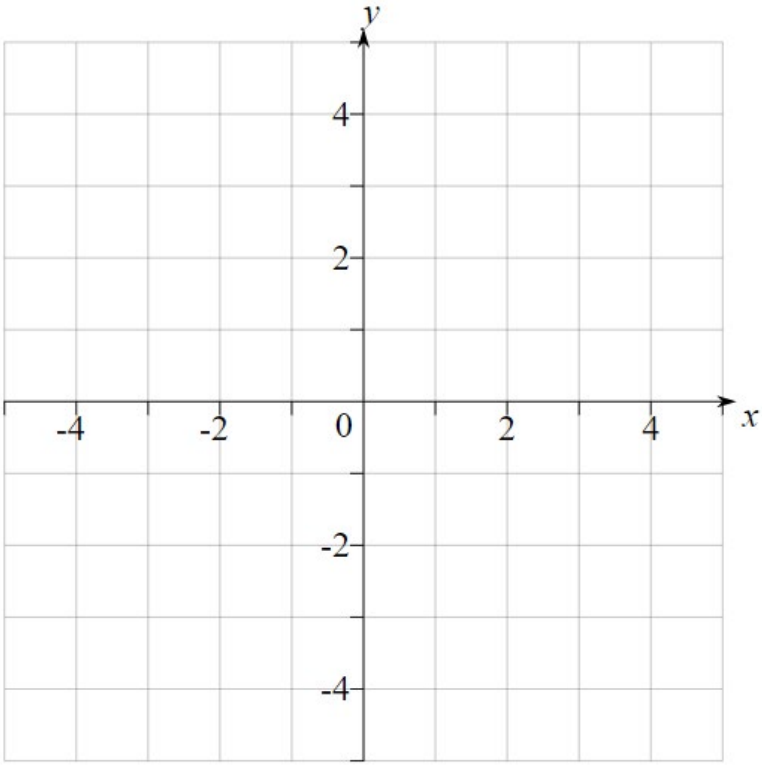
c. Hence, find the amount of salt which has flowed out of the tank over the first 25 minutes and the concentration of salt remaining in the tank after 25 minutes.

2 marks

Question 10 (4 marks)

Sketch the graph of $y = \dfrac{x^2 - 1}{x^2 - 4}$ on the axes below. Label any axis intercepts and turning points with their coordinates, and any asymptotes with their equation.



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End of Question and Answer Book

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